Detection and Localization of Appearance Faults in Reversible Circuits

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***Abstract*— This work presents a testing scheme for detection of two new type of faults (*gate appearance* and *control appearance fault*) in reversible circuit. The testing scheme not only efficiently detects the specified faults with minimum number of test vectors but localizes it simultaneously. Our developed approach only requires a single test vector to detect *gate appearance fault* while to find *control appearance fault* it needs *n* test vectors, where *n* is the number of input lines present in the circuit.**
The proposed technique can also work for very large circuits as well. In way to verify logical correctness of our developed technique, we have successfully tested different types of benchmark circuits over our proposed algorithms and obtained results are given at the end of this work.

Keywords: *-Reversible logic; appearance fault; test vector; missing gate fault*

I. Introduction

In the recent years, quantum computing [1-2] has come out as one of the most promising field of research due to its diversified applications in premier technologies like nanotechnology, cryptography and bioinformatics. At the same time remarkable progresses have been seen in this filed. For example, implementation of 2-qubit quantum gate in silicon [3], physical implementation of quantum circuits [4-6], even full fledge implementation of quantum computer [7] with lower qubits has already initiated.
In the mean time, the necessity of automated algorithms for designing quantum circuits is observed and to address this specific requirement the concept of reversible logic synthesis has evolved. Approaches like ESOP [8], BDD [9], Reed-Muller [10] are very well known synthesis algorithms that are used in efficient realization of reversible circuit.
But a new problem originates while working with such type of circuits that how to deal with the situation when these circuits become faulty [11] (very common types of faults are Single Missing Gate Fault (*SMGF*), Partial Missing Gate Fault (*PMGF*), Multiple Missing Gate Fault (*MMGF*) and Repeated Gate Fault (*RGF*)) as the detection of faults seem very important. Not only detection but fault localization also becomes necessary in way to restore the faulty circuit to its original form.
Since past couple of years, several investigations have been made by various research groups in developing efficient test detection algorithms. Here we are mentioning some of them.
Whether deterministic or probabilistic approaches work better in detection of various faults in reversible circuit is investigated in [12]. Construction of efficient test sets using integer linear program *(ILP)* for the detection of multiple stuck-at faults in reversible circuit is reported in [13]. Furthermore, in way to perform testing with lesser number of test vectors, a new approach based on optimal test sets (*OTS*) has introduced [14].
As the previous testing schemes were not able to give full fault coverage, a new testing mechanism (known as Ping Pong Test') with 100% fault coverage is introduced in [15].
But some questions arise here as several critics start claiming that full fault coverage in the reversible circuits is very expensive and time consuming. To counter this argument, in [16] theoretically it was proved that fault localization in the reversible circuits is easier than that in the classical circuits.
In way to address several concerns and aspects, different testing and fault detection mechanisms have been introduced in later years. Like, fault detection in reversible circuit using automated test patterns [17], universal testing technique for detection of all kinds of missing gate faults [18], testing without test vectors and only by employing boolean difference method [19].
But any of these testing schemes mentioned above always focuses in two aspects that the generated test sets to detect faults are smaller as much as possible and it should be scalable to test very large benchmarks. It has been seen that in most of the cases the first objective is reached but the second objective is not followed.
Apart from the common types of faults in reversible circuits there may exist some new type of faults like appearance of unwanted gates or control nodes in circuit and here in this work we have developed an algorithm to detect such type of faults. Our approach also satisfies the two aspects that it can test circuits with minimum number of test vectors and secondly it is capable to test very large benchmarks for finding possible faults.
The rest of the work is organized as follows. Preliminaries on reversible circuit and its related fault models are given in Section II. The proposed technique is explained in Section III. Experimental results obtained in our work are presented in Section IV and finally, the concluding remarks are given in Section V.

II. Background

Here we are introducing very preliminaries on reversible logic and associated fault models. Along with that we also have talked on two new types of faults - *gate appearance* and *control appearance faults* and have explained them with

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examples.

Definition 1: *A circuit can be called a "Reversible Circuit" if it satisfies the three basic criteria -1) there exist one to one mapping between input to output vectors and vice-versa 2)it should be fan-out free 3) and finally the circuit should consist of reversible gates only.*

The example of reversible gates like Feymen [20], Toffoli [21], Fredkin [22] are very well known gates that are used in designing reversible circuits.

The structure of a 2-control reversible gate (Toffoli gate) is shown in Fig. 1(a) and when such gates are cascaded over a set of control lines – a reversible circuit is formed. A reversible circuit designed with 4 such reversible gates over three control lines is shown in Fig. 1(b).

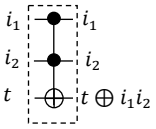

Fig. 1(a): 2-Control Toffoli gate

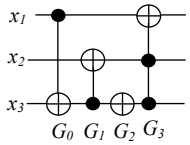

Fig. 1(b): Reversible circuit designed with 4 reversible gates

But a circuit consist of such gates may incur faults and a circuit can be called a faulty one if its output functionality does not confronts with its original behavior.

Generally, faults in reversible circuits originate in two parts – either in gates or in control nodes. Depending on the nature of errors in circuits, faults are broadly classified into four types and they are:

1. Single Missing Gate Fault (*SMGF*),
2. Partial Missing Gate Fault (*PMGF*),
3. Multiple Missing Gate Fault (*MMGF*) and
4. Repeated Gate Fault (*RGF*)

There are various ways to detect such type faults but most common of them is by applying test vectors. Basically, test vectors are Boolean patterns that compare the output string with some specified results and if it mismatches then first it confirms the presence of fault and in later time it starts locating the position of that faulty zone.

Here we are introducing some of the faults that we have mentioned earlier.

Definition 2: *If an entire gate disappears from a circuit by infusing fault in the design then such type of fault is known as Single Missing-Gate Fault (SMGF).*

See the design of Fig. 2(a) and if we assume that the first gate (marked by the dotted box) of that circuit goes missing from the design then the test vector <011> when applied over that circuit detect the fault as it find the output string <100> in spite of expected string <011>.

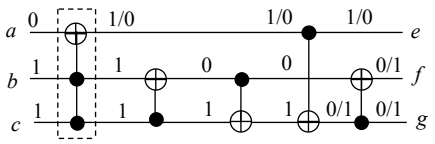

Fig. 2(a): *SMGF* in *ham3ndesign#1*

Definition 3: *MMGF is the super class SMGF and like as SMGF if more than one consecutive gate disappears from the circuit, then the fault type is termed as multiple missing-gate faults (MMGF).*

Example on *MMGF* is given in Fig. 2(b), where first two gates (marked by the dotted box) have disappeared from the circuit and the applied test vector <011> detects the *MMGF* in that design.

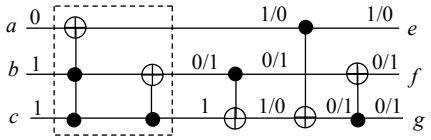

Fig. 2(b): *MMGF* in *ham3ndesign#1*

Apart from these two well known gate faults, a circuit may become faulty if some unwanted control nodes appear or disappear from the circuit and such fault is known as crosspoint fault [23].

Crosspoint fault can be of two types 1) *control disappearance fault* and 2) *control appearance fault*.

The faults due to *PMGF* belong to the category of *control disappearance fault*.

Definition 4: *If a circuit became faulty due to missing of control nodes then such type of fault is known as Partial missing gate fault (PMGF).*

For example, a control node (marked by the dotted box) in Fig. 2(c) goes missing by infusing fault in that design and such type fault belongs to the category of *PMGF*. In way to confirm the presence of fault, when we apply test vector <001> then it successfully detects this missing of control node in that circuit.

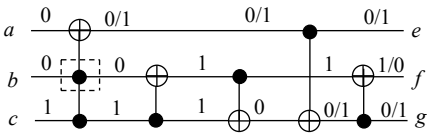

Fig. 2(c): *PMGF* in *ham3\design#1*

Definition 5: *If a fault in a circuit is caused due to sudden appearances of control nodes or gates then such type fault is known as control/gate appearance fault.*

For an example, when a control appears (see Fig. 3 (b)) in line x_2 of gate G_0 in the design of Fig. 3 (a) then the circuit become faulty and such type fault belongs to the category of *control appearance fault.* This injected fault can be detected if at least one of the extra control points has logic value 0, while the other control points are at logic 1.

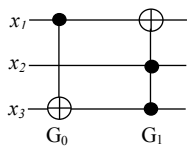

Fig. 3(a): Circuit before the occurrence of appearance fault

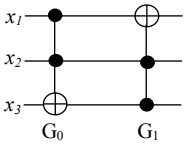

Fig. 3(b): Circuit after the occurrence of appearance fault

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Similarly, when a gate appears in between any of the gates present in the design then said fault is termed as *gate appearance fault.*

For example, consider the circuit of Fig. 4(a) that contains two gates over two circuit lines. Now, as any reversible circuit with n input lines may contain $n.2^{n-1}$ number of different gates, so if an n qubit circuit becomes faulty due to single *gate appearance fault* then any of the $n.2^{n-1}$ gates should appear in between any two circuit constituent gates.

Similarly, the design of Fig. 4(a) may contain 4 different gates (see Fig. 4(b)) in between the two existing gates by causing *gate appearance fault* in the circuit. We have shown the four different designs in Fig. 5, when any of the four gates appear in the original design.

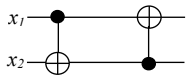

Fig. 4 (a): Reversible circuit with 2 input lines

Fig. 4 (b): Four possible gates with 2 input lines

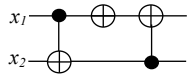

Fig. 5(a): *NOT* gate appears in line x_1 of Fig. 4(a)

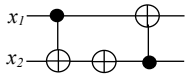

Fig. 5(b): *NOT* gate appears in line x_2 of Fig. 4(a)

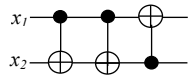

Fig. 5(d): Appearance of *CNOT* in the design of Fig. 4(a)

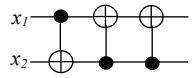

Fig. 5(e): Appearance of *CNOT* in the design of Fig. 4(a)

III. PROPOSED TECHNIQUE

Here, we are stating the mechanism for detection of unwanted gates and controls leading to appearance faults in reversible circuits. The developed testing scheme is completed in three phases and they are 1) *circuit augmentation* 2) *fault detection* and 3) *fault localization phase*.

For easy of understanding, we are explaining the entire procedure with an example. Let consider the circuit of Fig. 6(a) as the input circuit which initially does not have any faults and in later stages we will inject faults in that circuit which are detected and localized using our developed testing algorithm.

A. *Phase1: Circuit Augmentation Phase:*

In this augmentation phase, we transform an input circuit to the testable circuit by adding a set of gates and one additional line to the circuit. There are two steps involved in transforming an input circuit to augmented circuit (also called testable circuit).

Step1: An additional line (A_L) is added at the bottom of the input circuit and we declare the line A_L as a constant line whose value is set to 1.

Step2: For each of the *Toffoli* gates present in the input circuit, an additional copy gate is placed in the circuit in such a way that a single control from all the copy gates pass through the line A_L.

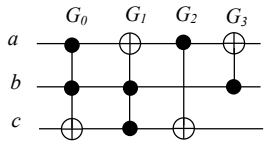

Fig. 6(a): Input circuit (C_{in})

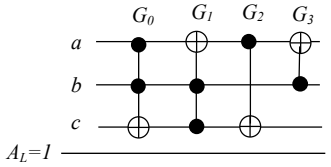

Fig. 6(b): Circuit obtained after executing *step1* of *phase1*

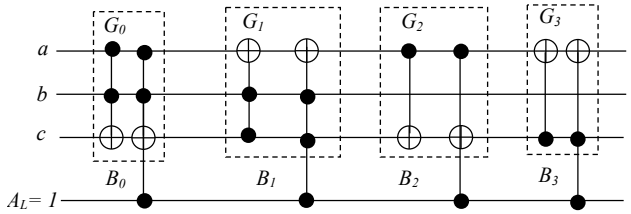

Fig. 6(c): Modified Circuit after *step2* of *phase1*

Now, the input circuit of Fig. 6(a) is converted into the testable circuit as shown in Fig. 6(c) and if A_L is set to 0 then the testable circuit will be same as input circuit of Fig. 6(a).

Now, we divide the testable circuit into number of blocks, where each of the blocks contains an original gate and its copy gate. So, if a circuit has n number of gates then there will be created n number of blocks and an important design aspect related to it is that, the input patterns applied to any of the blocks remain same at the output segment of that block though after passing the patterns through the gates present in that block.

See Fig. 6(d), where the testable circuit of Fig. 6(c) is divided into four blocks namely B_0, B_1, B_2 and B_3. The figure of 6(d), 6(e) and 6(f), show how the patterns applied to the input of these blocks remain same at end of the blocks.

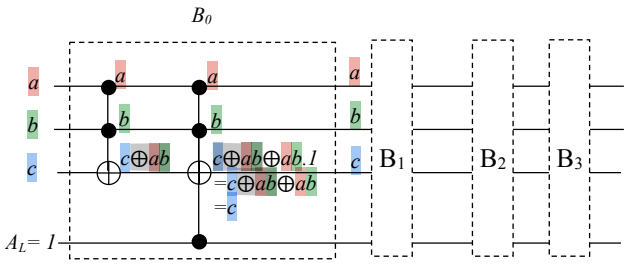

Fig. 6(d): Block B_0 of Fig. 6(c) produces equal input and output

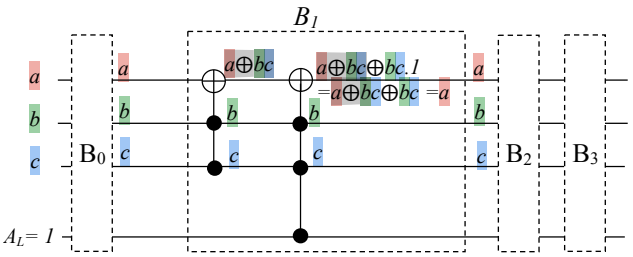

Fig. 6(e): Block B_1 of Fig. 6(c) produces equal input and output

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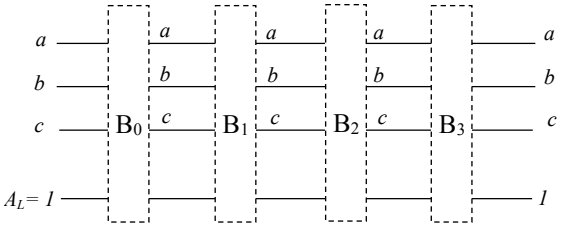

Fig. 6(f): Each block of Fig. 6(c) produces equal input and output

B. *Phase2: Fault detection phase:*

In this phase, a circuit is examined to determine whether it is faulty or fault free. Such faults can be detected by applying test vectors. Such test vectors contain a specific boolean patterns, when they are applied over a certain circuit it checks the circuit correctness by judging the applied input pattern and the obtained output pattern. If the desired output is obtained for the given test pattern then we can say that the given circuit is fault-free; otherwise, there is defect in the given circuit. So, production of proper test patterns is necessary to test a circuit.
Here we are stating what types of test patterns are to be generated and how the fault detection process works to find errors.

B.1: Test vector generation:

B.1.1: *For gate appearance fault:*

If it assumed that a new Toffoli gate appears in between any two gates then as a consequence the circuit's original functionality changes by causing a fault in the design.
In such a case, we have seen that for any reversible circuit with *m* number of gates over *n* input lines requires the test pattern <11…1>, to find out the said error.
The reason for choosing such a test pattern is that as we have decomposed the entire circuit into multiple blocks that never alters the applied test vector unless any unwanted gates appear in between. In that situation when an all 1 test vector is applied then due to appearance of a new gate the applied test vector changes (all 1s activates all the controls of the new gate resulting change in the test vector) which finally mismatches with the input all 1 pattern by confirming the presence of fault.

B.1.2: *For control appearance fault:*

The fault in a circuit due to appearance of a single control can be found with a set of test patterns and it is found that for a circuit with *m* input lines requires *m* different test vectors to detect any *control appearance fault*.
For example, a test vector of form <011…11> can find any control appearance fault over 0th line and interestingly if the bit value 0 in previous pattern is shifted to its right (new pattern become <101…11>) then it can find fault over 1st line and in this way by right-shifting the position of bit 0 any control appearance fault over any of the lines can be tracked.

B.2: *Fault Detection Mechanism:*

To check whether a circuit faulty or not, we have to pass the previously generated test patterns through the circuit and if the output test patterns are same as the input test patterns, then the circuit is fault free, otherwise it is a faulty one.

For example, let us consider in the circuit of Fig. 6(c) an unwanted gate G_P appears (after appending the new gate in Fig. 6(c), the modified circuit is shown in Fig. 7(a)) in between blocks B_1 and B_2 by causing a *gate appearance fault*.
Now, when the test pattern <111> is applied over the circuit in Fig. 7(a), then the gate G_P changes the input vector <111> to <101> resulting different input-output pair which confirms the presence of an error in that circuit.
Again if we consider that in spite of appearing a new gate in the circuit, a control appears in gate G_2 of block B_2 by causing a control appearance fault (see Fig. 7(b)).
As like previous situation, when the test pattern <101> is applied over the circuit, then the output changes to <100> that differs with the input pattern which again confirm the presence of fault.
So, here we can see that our generated test vectors are very sufficient to find any of the control and gate appearance fault in a circuit if exist and now we will show how fault localization is performed by finding its fault's origination point. The localization of a fault is very important in way to restore the circuit's functionality to its original form.

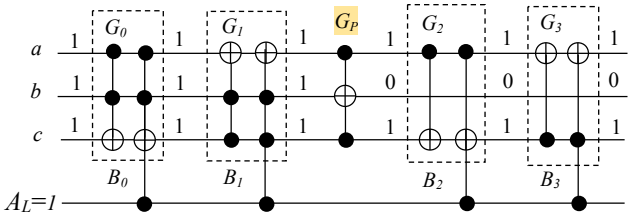

Fig. 7(a): Faulty circuit due to appearance of gate G_P in between gate G_1 and G_2

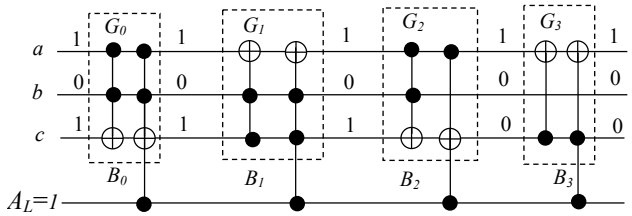

Fig. 7(b): Faulty circuit due to appearance of additional control node in gate G_2 over line *b*

C. *Fault localization phase:*

Previously, we have stated that our testable design is segmented in several blocks and here that segmentation will help us in locating the fault.
If we find that the applied input test vector differs with the output vector of the circuit then it confirms the presence of control appearance fault in the design. In that situation, we scan the entire circuit from input section to output section and perform block wise checking. If we find the applied input-output test vector is different in a certain block then it is confirmed that the block contains the error.
Similarly, if we find that output vector of a block (B_i) is different from the input vector of the immediate block (B_{i+1}) then we can ascertain that there is a gate appearance fault in that circuit and the fault is in between block B_i and B_{i+1}.
For example, when there was gate appearance fault in the circuit of Fig. 7(a) due appearance of the gate G_P, the fault is tracked in between block of B_1 and B_2 by finding that the output vector (<111>) from block B_1 is different with input

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vector (<101>) of B_2.
Similarly, the control appearance fault of Fig. 7(b) is tracked to block B_2.

IV. Experimental Results

We have applied our algorithm over several benchmark circuits [24] and obtained results are shown in TABLE-I. As could not find any related works to which we can compare our results, we have only given our obtained results in support of the proposed approach.
In the result table, column 1 shows benchmark circuit names; column 2 and column 3 denote the number of input line (n) and number of gates (N) respectively. Column 4 and column 5 of TABLE-I represents possible number of *gate appearance faults* (N_{GA}) and *control appearance faults* (N_{CA}) in the respective benchmark circuits. Total numbers of test vectors required for detecting *gate appearance fault* (T_{GA}) and *control appearance fault* (T_{CA}) are shown in column 6 and 7, respectively.

V. Conclusion

This work has presented a testing scheme to test two new types of faults - *gate appearance* and *control appearance* fault in reversible circuit. We have tested the reversible circuit for possible *gate appearance* fault with minimum number of test vectors and also have shown how efficiently gate appearance fault can be detected by making a suitable testable design. For the both type of faults, we not only have shown the fault detection mechanism but also have successfully located the origination of fault in a circuit. Our developed scheme is also applicable for very large size benchmarks as well.

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Table-I: Experimental results from the proposed fault testing algorithm

Benchmark names	***Size (n)***	***No. of gates (G)***	***No. of Gate appearance faults (N_{GA})***	***No. of control appearance faults (N_{CA})***	***No. of test vectors required in detecting gate appearance fault (T_{GA})***	***No. of test vectors required in detecting control appearance fault (T_{CA})***
317tc	3	6	12	5	1	3
ham3tc	3	5	12	4	1	3
hwb4d1	4	17	32	24	1	4
mod5d2	5	9	80	25	1	5
mod5d1	5	8	80	20	1	5
5mod5d1	6	17	192	47	1	6
mod5adder	6	21	192	78	1	6
rd53d1	7	12	448	44	1	7
2of5d2	7	12	448	53	1	7
ham7d1	7	23	448	104	1	7
rd53d2	8	12	1024	64	1	8
6symd2	10	20	5120	147	1	10
rd73d2	10	20	5120	145	1	10
9symd2	12	28	24576	259	1	12
rd84d1	15	28	245760	344	1	15
cm150a	21	53	22020096	276	1	21
frg1	28	212	3758096384	2540	1	28
apex2	39	1746	$10*2^{12}$ (approx.)	25912	1	39
seq	41	5990	$4*2^{13}$ (approx.)	41996	1	41
e64	65	387	$11*2^{18}$ (approx.)	2080	1	65
apex5	117	2909	$8*2^{34}$ (approx.)	136453	1	117
frg2	143	3724	$7*2^{44}$ (approx.)	593725	1	143

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